SILAS KWABLA GAH

PhD Candidate in Computer Science · Deep Learning, World Models & Test-Time Adaptation

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Research Profile

Final-year PhD candidate in Computer Vision and Machine Learning at the University of Ghana, working on geometric 3D scene understanding, multimodal foundation models, test-time adaptation ($\Delta\theta = 0$), and reinforcement learning for grounded visual reasoning. Research focuses on inference-time coordination of independently operating visual models, camera-geometry-based 2D–3D lifting, semantic evidence preservation, and cross-view consistency as self-supervised and process-level oversight signals. Experienced with large-scale 3D/video point-cloud evaluation, PyTorch GPU acceleration, and memory-bounded inference pipelines.

Education

PhD in Computer Vision and Machine Learning — Expected defence: October 2026
University of Ghana, Legon — *Supervisor: Prof. Ebenezer Owusu*

Thesis: *Visual Perception under Low Supervision: Inference-Time Coordination of Frozen Foundation Model Representations*
Research investigates frozen foundation-model composition, open-vocabulary 2D/3D segmentation, semantic evidence preservation (avoiding premature collapse), multi-view geometric consensus, and reliability-aware inference.

MPhil in Computer Science — 2016
University of Ghana, Legon — *Accra, Ghana*

BSc in Computer Science — 2012
Ghana Institute of Management and Public Administration (GIMPA) — *Accra, Ghana*

HND in Electrical & Electronic Engineering — 2009
Accra Technical University — *Accra, Ghana*

Research Experience

PhD Research — 3D Scene Understanding, Foundation Models & Test-Time Coordination University of Ghana

- Formulated **Reliability-Calibrated Adaptive Semantic Arbitration (RCASA)**, an inference-time framework that coordinates heterogeneous foundation models without gradient updates ($\Delta\theta = 0$), combining Platt calibration, uncertainty modeling, and contextual reliability weighting.
- Identified **premature semantic collapse** as a fundamental information bottleneck in multi-model composition; designed controlled Top-k retention experiments demonstrating that preserving secondary semantic hypotheses substantially improves multi-source coordination.
- Developed a camera-geometry-based **2D-to-3D lifting and consensus pipeline** using calibrated intrinsics, extrinsic poses, and multi-view semantic accumulation to reconcile dense 3D vision-language predictions (**RegionPLC**) with projected 2D promptable segmentations (**SAM3**).
- Evaluated across benchmark protocols on **ScanNet200** (312 scenes, ~50M points), **ScanNet++**, and alternative sparse sources (**GroundingDINO+SAM2.1**) using paired scene-level bootstrap confidence intervals.
- Built streaming and memory-bounded infrastructure for evaluating dense per-point semantic distributions via chunked processing, float16 caching, sparse tensors, and calibration diagnostics.

Multimodal RL — Cross-View Agreement as a Process-Level Reward — University of Ghana

- Developed a process-level grounding reward leveraging consensus between frozen 3D and image-based perception sources, converting cross-view physical consistency into self-supervision without human-annotated reward labels.
- Built a **GRPO/VERL** multimodal RL pipeline for **Qwen2.5-VL-7B-Instruct** with **vLLM** rollout generation and structured grounding rewards; completed reward-validation and counterfactual discrimination studies against specification gaming.
- Evaluated reward dynamics across 1,000+ multimodal reasoning rollouts, separating spatial grounding validity from final-answer correctness and eliminating area-bias gaming ($\rho > 0.72 \to 0.334$).

Selected Publications and Manuscripts

- **Beyond Argmax: A Mechanistic Study of Semantic Retention in Frozen Foundation-Model Composition for Generalized Few-Shot 3D Segmentation.**
  ICLR 2027 Conference Submission; preprint available. First Author.

- **Training-Free Open-Vocabulary 3D Point-Cloud Segmentation on the Generalized Few-Shot Benchmark.**
  ACCV 2026, Paper #482; preprint available. First Author.
- **Training-Free Generalised Few-Shot Segmentation through Open-Vocabulary Semantic Arbitration.**
  WACV 2027, Paper #167. First Author.
- **Cross-View Agreement as a Label-Free Process Reward for Multimodal Reasoning.**
  SACAIR 2026 submission. First Author.
- **Reliability-Calibrated Adaptive Semantic Arbitration of Frozen Foundation Model Representations.**
  Manuscript in preparation (targeted for IEEE TPAMI). First Author.

Technical Skills

Programming	Python, Java, C++
Deep Learning / RL	PyTorch, Transformers, vLLM, LoRA/QLoRA, SFT, GRPO, VERL
GPU & Systems	CUDA fundamentals, Triton kernel development and benchmarking, GPU profiling, memory-efficient inference and evaluation pipelines
3D & Geometric Vision	Point clouds, camera intrinsics/extrinsics, coordinate transforms, posed-view projection, 2D–3D lifting, multi-view semantic aggregation
Foundation Models	RegionPLC, SAM3, SAM2.1, GroundingDINO, CLIP, DINOv2, Qwen2.5-VL
Methodology	Paired bootstrap confidence intervals, calibration analysis (ECE, Brier), controlled ablations, matched causal interventions, large-scale evaluation

Teaching, Professional Experience & Service

Part-time Lecturer — Accra Technical University (2022 – Present)

- Lecturing in computing, algorithms, and applied machine learning; previously taught at GIMPA.

Founder & Lead Architect — EED Soft Consult (2018 – Present)

- Architected and delivered large-scale, production-grade distributed and mobile software systems for enterprise and public-sector clients in Ghana.

Academic Service — Reviewer, SACAIR 2026 Program Committee

Research Interests

World models; collaborative perception; test-time adaptation; self-supervised learning; 3D scene understanding; foundation-model composition; embodied AI; multimodal reinforcement learning.